
Software Requirements Specification

for

Automated Lecture Summarizer and Flashcard Generator

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements of the Automated Lecture Summarizer and Flashcard Generator system. This project aims to assist students in learning by automatically generating concise summaries and flashcards from lecture materials such as text notes, PDFs, or recorded lecture transcripts.

This SRS provides a detailed description of the system's features, interfaces, constraints, and expected behavior. It serves as a reference for developers, testers, and users throughout the software development lifecycle.

1.2 Document Conventions

This document follows the IEEE Std 830-1998 standard for Software Requirements Specifications. The following conventions apply: (1) Requirements are written using the keyword SHALL for mandatory requirements and SHOULD for recommended features. (2) Each functional requirement is assigned a unique identifier in the format FR-XX. (3) Non-functional requirements are prefixed with NFR-XX. (4) Section headings follow a hierarchical decimal numbering system. (5) Bold text is used to highlight critical terms and important definitions. (6) Higher-level requirements are assumed to propagate their priority levels to their child requirements unless stated otherwise.

1.3 Intended Audience and Reading Suggestions

This SRS document is intended for the following stakeholders:

- Developers: SHALL use this document to understand the functional and technical requirements for implementation.
- Project Guide / Faculty: SHALL use this document to evaluate project feasibility, design, and scope.
- Testers: SHALL use this document to verify whether the developed system satisfies all specified requirements.
- Students (End Users): SHOULD use this document to understand the purpose and functionalities of the system.
- Readers are recommended to begin with the Introduction section for an overview, proceed to the Overall Description for system context, and then review the detailed requirements in subsequent sections.

1.4 Product Scope

The Automated Lecture Summarizer and Flashcard Generator is an educational software system designed to transform lecture content into simplified study material.

The system SHALL accept lecture notes, text documents, PDF files, or transcripts as input. It SHALL analyze the content using Natural Language Processing (NLP) techniques and generate:

- A summarized version of the lecture content.
- Important keywords and concepts.
- Flashcards in question-answer format for revision purposes.

The system aims to improve student productivity, enhance learning efficiency, and support exam preparation through automated content analysis.

1.5 References

The following references were consulted during the preparation of this SRS:

1. IEEE Std 830-1998, "IEEE Recommended Practice for Software Requirements Specifications," IEEE, 1998.
2. Sommerville, I., "Software Engineering," 10th Edition, Pearson Education, 2015.
3. Pressman, R. S., "Software Engineering: A Practitioner's Approach," 8th Edition, McGraw-Hill, 2014.
4. Python Documentation, Python Software Foundation, <https://docs.python.org/3/>.
5. FastAPI Documentation, <https://fastapi.tiangolo.com/>.
6. MongoDB Documentation, MongoDB Inc., <https://www.postgresql.org/docs/>.
7. Hugging Face Transformers Documentation, <https://huggingface.co/docs/transformers/> (huggingface.co in Bing).
8. OpenAI Whisper Speech-to-Text Documentation, <https://platform.openai.com/docs/guides/speech-to-text> (platform.openai.com in Bing).
9. ReactJS Documentation, Meta Platforms, <https://react.dev/>.
10. Natural Language Processing research papers and educational resources related to text summarization and flashcard generation.

2. Overall Description

2.1 Product Perspective

- FR-01: The Automated Lecture Summarizer and Flashcard Generator SHALL be a new, self-contained educational tool designed to process lecture content efficiently.
- FR-02: The system SHALL integrate speech-to-text transcription, natural language processing (NLP) summarization, and flashcard generation into a single platform.
- FR-03: The product SHALL accept lecture videos or audio files, convert them into text, summarize the content, and generate interactive flashcards.
- FR-04: The system SHOULD provide APIs for integration with external Learning Management Systems (LMS).

2.2 Product Functions

At a high level, the system will provide the following functions:

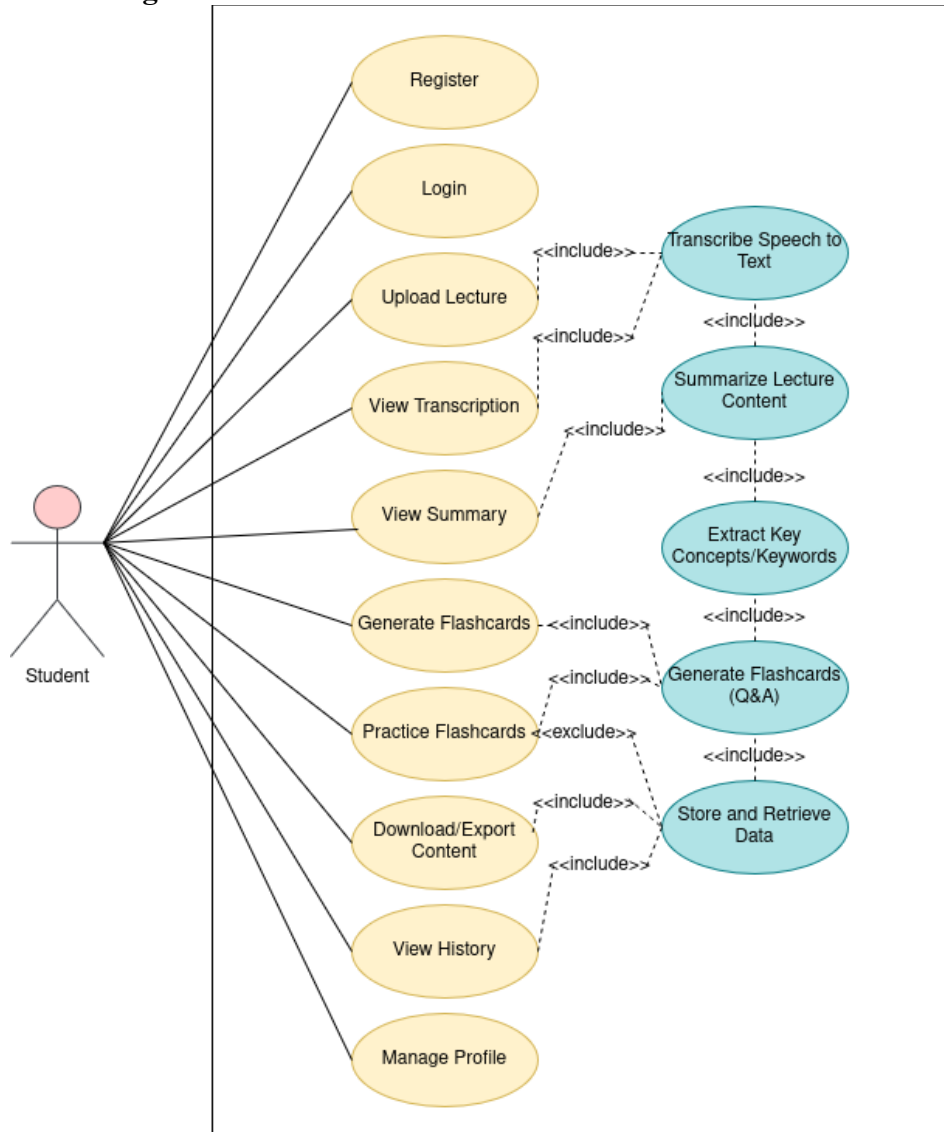
- FR-05: The system SHALL allow users to upload lecture videos or audio files.
- FR-06: The system SHALL perform speech-to-text conversion to generate transcripts.
- FR-07: The system SHALL generate concise summaries of lecture transcripts using NLP models.
- FR-08: The system SHALL automatically create Q&A flashcards from summarized content.
- FR-09: The system SHALL provide a user interface for viewing summaries and practicing flashcards.
- FR-10: The system SHALL store transcripts, summaries, and flashcards in a database for retrieval.

2.3 User Classes and Characteristics

The system is intended for the following user classes:

- FR-11: The system SHALL support students as primary users, who will upload lectures, view summaries, and practice flashcards.

Use Case Diagram:



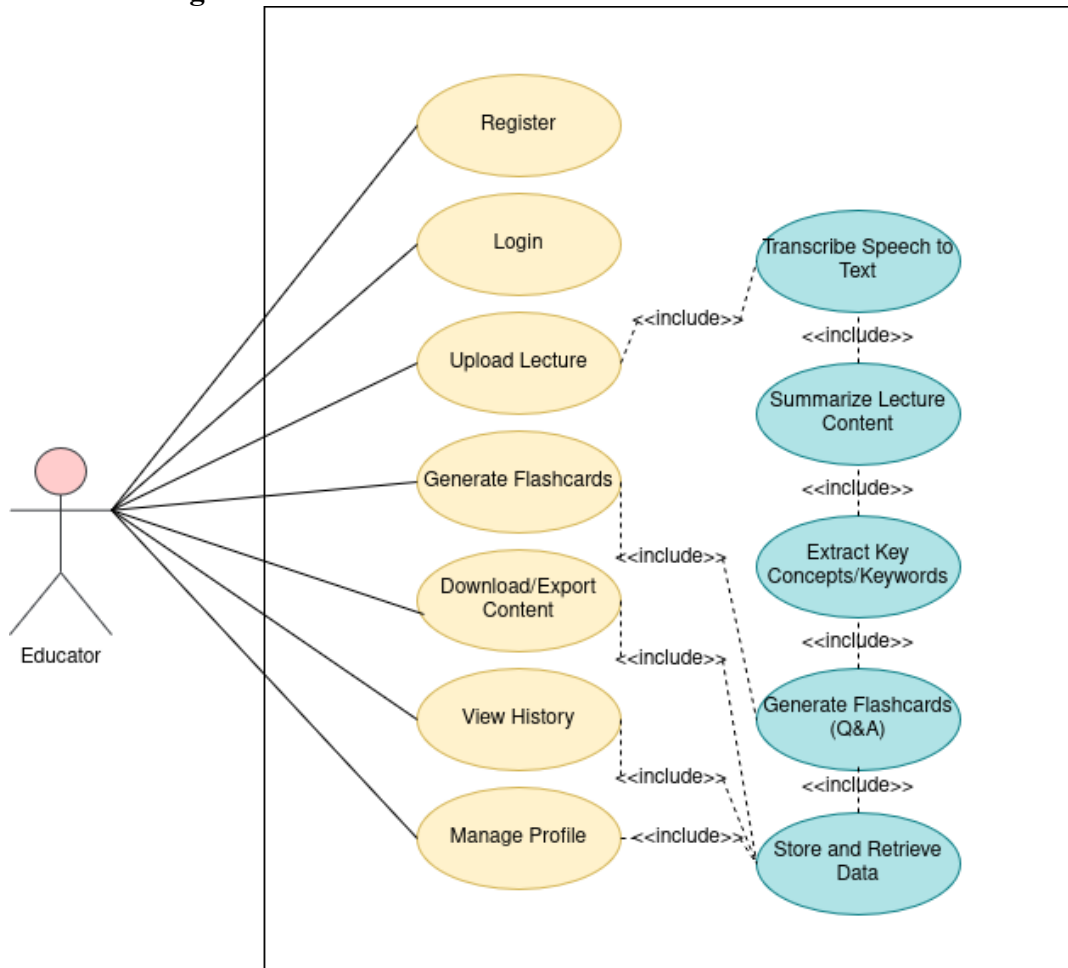
Use Case Description

- **Use Case Name:** Upload Lecture
- **Primary Actor:** Student
- **Secondary Actor:** Administrator
- **Description:** Allows the student to upload an audio/video/PDF lecture file to the system to initiate automatic transcription, summarization, and keyword extraction.

- **Precondition:** Student must be logged into the system.
- **Postcondition:** The lecture file is successfully uploaded and stored. The system triggers the processing pipeline.
- **Trigger:** The student clicks the "Upload Lecture" button.
- **Main Success Scenario**
 - Student selects the "Upload Lecture" option from the dashboard.
 - System prompts the student to select a valid audio/video/PDF file from their device.
 - Student selects the file and clicks "Submit".
 - System validates the file format and size.
 - System uploads the file and securely stores it.
 - System automatically initiates the transcription process.
 - System automatically initiates the summarization process.
 - System extracts key concepts.
 - System displays a success message to the student.
- **Exception Scenario**
 - Invalid File Format/Size
 - Network Interruption

- FR-12: The system SHALL support educators as secondary users, who may upload lecture content and share flashcards.

Use Case Diagram:



Use Case Description

- **Use Case Name:** Generate Flashcards
- **Primary Actor:** Educator
- **Secondary Actor:** Administrator
- **Description:** Enables an educator to generate AI-based flashcards from uploaded lecture content or summaries. The system extracts key concepts and creates question-answer flashcards to support effective learning and revision.
- **Precondition:** Educator is logged in and lecture content is available.
- **Postcondition:** Flashcards are generated and saved for viewing or sharing.
- **Trigger:** Educator selects Generate Flashcards.

- **Main Success Scenario**

Educator logs in.

Opens a lecture or summary.

Clicks Generate Flashcards.

System analyzes the content.

Flashcards are generated.

Educator reviews and saves/shares them.

- **Exception Scenario**

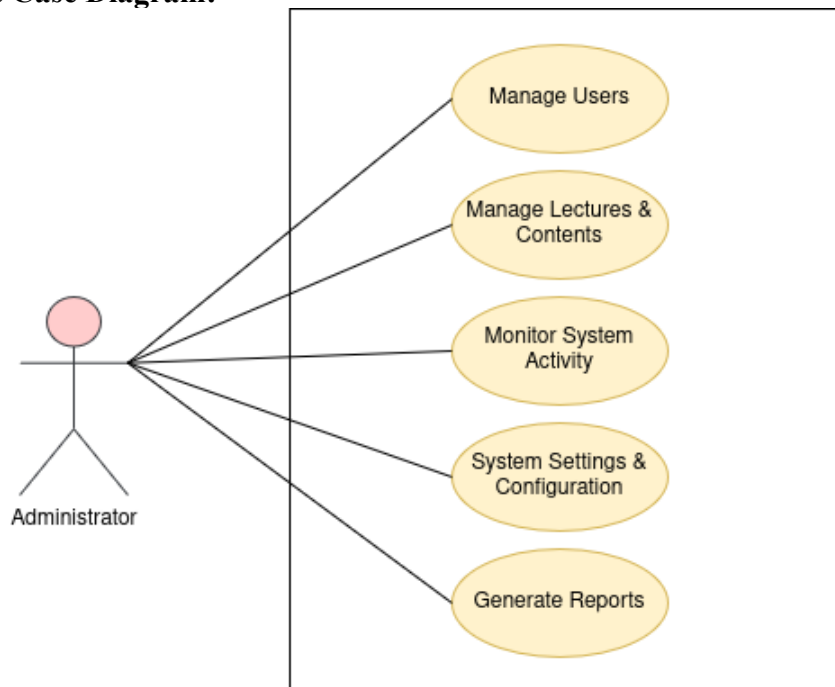
Lecture content is unavailable.

Processing fails.

Flashcard generation cannot be completed.

- FR-13: The system SHALL support administrators, who will manage user accounts, monitor system performance, and ensure data security.

Use Case Diagram:



Use Case Description

- **Use Case Name:** Manage User Accounts
- **Primary Actor:** Administrator
- **Secondary Actor:** System

- **Description:** Allows the administrator to manage user accounts within the system. The administrator can add, update, deactivate, or remove student and educator accounts to ensure secure and efficient system administration.
- **Precondition:** Administrator is authenticated with admin privileges.
- **Postcondition:** User account information is updated successfully.
- **Trigger:** Administrator selects Manage Users.
- **Main Success Scenario**
Administrator logs in.
Opens the user management page.
Views the list of users.
Adds, edits, or removes a user.
System validates the request.
Changes are saved successfully.
- **Exception Scenario**
User not found.
Insufficient permissions.
Database update fails.

2.4 Operating Environment

The Automated Lecture Summarizer and Flashcard Generator shall operate in the following environment:

- FR-14: The client side SHALL operate on modern web browsers (Google Chrome v100+, Mozilla Firefox v100+, Microsoft Edge v100+, Safari v15+) running on Windows 10/11, macOS 12+, or Ubuntu 20.04+.
- FR-15: The server side SHALL use Python 3.11+ with FastAPI framework deployed on a Linux-based server (Ubuntu 22.04 LTS recommended).
- FR-16: The database SHALL be MongoDB Atlas, a cloud-hosted NoSQL database service, accessed through a secure connection from the FastAPI backend.
- FR-17: The frontend SHALL be React.js v18+ built with Vite.
- FR-18: The system SHALL require a stable internet or LAN connection with minimum 10 Mbps bandwidth.
- FR-19: The system SHALL integrate Whisper (OpenAI) for transcription of lecture audio.
- NFR-01: All client-server communication SHALL be secured using HTTPS with TLS 1.2+.

2.5 Design and Implementation Constraints

- NFR-02: The system SHALL comply with applicable data privacy regulations (e.g., GDPR).
- NFR-03: The system SHALL require internet connectivity for cloud-based transcription and summarization.
- NFR-04: The system SHALL support scalability to handle multiple users simultaneously.
- NFR-05: The system SHOULD utilize open-source NLP models and APIs where feasible.

2.6 User Documentation

The following documentation will be provided:

1. FR-20: The system SHALL provide a user manual with instructions for uploading lectures, viewing summaries, and practicing flashcards.
2. FR-21: The system SHALL provide online help integrated into the user interface.
3. FR-22: The system SHALL provide tutorials for first-time users.
4. FR-23: The system SHOULD provide API documentation for integration with external systems.

2.7 Assumptions and Dependencies

1. NFR-06: It is assumed that users SHALL have access to stable internet connections.
2. NFR-07: The system SHALL depend on third-party speech-to-text services (e.g., Whisper, Google Speech API).
3. NFR-08: It is assumed that lecture audio SHALL be of reasonable quality with minimal background noise.
4. NFR-09: The system SHALL depend on cloud storage services for video handling.
5. NFR-10: It is assumed that users SHALL primarily interact via web browsers or mobile applications.

3. External Interface Requirements

3.1 User Interfaces

The system SHALL provide a web-based user interface accessible through modern web browsers.

Student Interface

- The student SHALL be provided with a login interface for secure authentication.
- The dashboard SHALL provide options to upload lecture files, view transcripts, view summaries, access flashcards, and track previously processed lectures.
- The upload interface SHALL allow users to select supported audio, video, PDF, or text files.
- The system SHALL display upload progress and processing status.
- The summary interface SHALL display the generated summary in a clear and readable format.
- Important keywords and concepts SHALL be displayed separately from the summary.
- The flashcard interface SHALL display questions and answers interactively.
- The interface SHOULD allow students to reveal answers, move between flashcards, and restart a flashcard session.
- Appropriate success, warning, and error messages SHALL be displayed for user actions.

Educator Interface

- Educators SHALL have access to uploaded lecture materials and generated summaries.
- The interface SHALL provide an option to generate flashcards from lecture content.
- Educators SHOULD be able to review, save, and share generated flashcards.

Administrator Interface

- Administrators SHALL have access to a user-management interface.
- The administrator interface SHALL provide options to view, add, update, deactivate, and remove user accounts.

- The system SHALL display appropriate confirmation messages for administrative operations. The user interface SHALL maintain consistent navigation, buttons, labels, error messages, and page layouts throughout the application.

3.2 Hardware Interfaces

- The system SHALL primarily operate as a software-based web application and SHALL not require dedicated hardware.
- The client SHALL operate on computers, laptops, tablets, or smartphones capable of running a supported web browser.
- The client device SHALL provide sufficient storage for temporarily selecting lecture files for upload.
- For recorded lectures, the input audio/video SHALL be obtained from the user's device storage.
- The server SHALL provide sufficient CPU, memory, and storage resources for transcription, NLP processing, database operations, and file handling.
- No specialized sensors or hardware devices SHALL be required.
- The system MAY use cloud infrastructure for computationally intensive processing and storage.

3.3 Software Interfaces

The system SHALL interact with the following software components:

Software Component	Version	Purpose
React.js	18+	Development of the web-based frontend
Vite	Latest compatible version	Frontend development and build tool
Python	3.11+	Backend and NLP processing
FastAPI	Compatible version	REST API development
MongoDB	Atlas	Cloud-based Persistent data storage
Whisper	Compatible version	Speech-to-text transcription
Hugging Face Transformers	Compatible version	NLP-based summarization and text processing
Linux/Ubuntu	Ubuntu 22.04 LTS recommended	Server operating environment

- The frontend SHALL communicate with the FastAPI backend through RESTful APIs.
- The backend SHALL communicate with MongoDB Atlas for storing user information, lecture metadata, transcripts, summaries, keywords, and flashcards.
- The speech-to-text component SHALL receive audio extracted from uploaded lecture files and return a text transcript.

- The NLP component SHALL receive lecture transcripts and produce summaries, keywords, and content suitable for flashcard generation.

3.4 Communications Interfaces

- The client and server SHALL communicate using HTTP/HTTPS.
- All production client-server communication SHALL use HTTPS with TLS 1.2 or higher.
- REST APIs SHALL be used for communication between the React frontend and FastAPI backend.
- Data exchanged between the frontend and backend SHOULD use JSON format unless file transfer is required.
- Lecture files SHALL be transferred using secure HTTP multipart/form-data requests.
- Authentication information SHALL be transmitted securely.
- The system SHALL validate incoming API requests before processing them.
- The server SHALL return appropriate HTTP status codes for successful requests, client errors, authentication failures, and server errors.
- The system SHALL support a minimum network bandwidth of 10 Mbps as specified in the operating environment.

4. System Features

4.1 User Authentication and Account Management

4.1.1 Description and Priority

This feature provides secure authentication and account management for students, educators, and administrators.

Priority: High

4.1.2 Stimulus/Response Sequences

User opens the application.

System displays the login page.

User enters valid credentials.

System verifies the credentials.

System identifies the user's role.

System redirects the user to the appropriate dashboard.

4.1.3 Functional Requirements

- FR-24: The system SHALL provide a login mechanism for registered users.
- FR-25: The system SHALL authenticate users before providing access to protected resources.
- FR-26: The system SHALL identify users according to their assigned role.
- FR-27: The system SHALL provide separate access privileges for students, educators, and administrators.

- FR-28: The system SHALL reject invalid authentication credentials.
- FR-29: The system SHALL allow authenticated users to log out securely.
- FR-30: The system SHALL prevent unauthenticated users from accessing protected application functions.

4.2 Lecture Upload and Processing

4.2.1 Description and Priority

This feature allows students and educators to upload lecture content for automatic processing.
Priority: High

4.2.2 Stimulus/Response Sequences

User selects Upload Lecture.
System displays the file-selection interface.
User selects a supported lecture file.
System validates the file type and size.
System uploads and stores the file.
System starts the processing pipeline.
Audio/video content is transcribed when required.
The transcript is sent for summarization.
Key concepts are extracted.
Processing status is displayed to the user.

4.2.3 Functional Requirements

- FR-31: The system SHALL allow authorized users to upload lecture audio, video, PDF, and text files.
- FR-32: The system SHALL validate uploaded file types.
- FR-33: The system SHALL validate uploaded file sizes.
- FR-34: The system SHALL securely store uploaded lecture files.
- FR-35: The system SHALL initiate processing after successful upload.
- FR-36: The system SHALL display the processing status to the user.
- FR-37: The system SHALL display an appropriate error message when an upload fails.
- FR-38: The system SHALL prevent unsupported or invalid files from entering the processing pipeline.

4.3 Speech-to-Text Transcription

4.3.1 Description and Priority

This feature converts spoken content from lecture audio or video into machine-readable text.
Priority: High

4.3.2 Stimulus/Response Sequences

User uploads an audio/video lecture.
System extracts or accesses the audio stream.
Whisper processes the audio.
The speech is converted into text.
System stores the generated transcript.
The transcript is made available for summarization.

4.3.3 Functional Requirements

- FR-39: The system SHALL convert supported lecture audio into text using Whisper.
- FR-40: The system SHALL associate each generated transcript with its corresponding lecture.
- FR-41: The system SHALL store successfully generated transcripts in the database.
- FR-42: The system SHALL report transcription failures to the user.
- FR-43: The system SHOULD preserve the generated transcript without unnecessary modification.
- FR-44: The system SHALL pass the generated transcript to the summarization module.

4.4 Automatic Lecture Summarization

4.4.1 Description and Priority

This feature automatically converts lengthy lecture transcripts into concise and meaningful summaries using NLP techniques.
Priority: High

4.4.2 Stimulus/Response Sequences

A transcript becomes available.
System sends the transcript to the NLP summarization module.
NLP model analyzes the transcript.
Important information is identified.
A concise summary is generated.
The summary is stored in the database.
The summary is displayed to the user.

4.4.3 Functional Requirements

- FR-45: The system SHALL generate a summary from a valid lecture transcript.
- FR-46: The system SHALL use NLP techniques or models for summarization.
- FR-47: The generated summary SHALL retain the important concepts of the lecture.
- FR-48: The system SHALL store generated summaries for future retrieval.
- FR-49: The system SHALL display the generated summary through the user interface.
- FR-50: The system SHALL notify the user if summarization fails.
- FR-51: The system SHOULD generate summaries that are substantially shorter than the original lecture transcript.

4.5 Keyword and Key Concept Extraction

4.5.1 Description and Priority

This feature identifies important keywords and concepts from lecture content to help students quickly understand the major topics.

Priority: Medium

4.5.2 Stimulus/Response Sequences

System receives the lecture transcript or processed content.

NLP processing identifies important terms and concepts.

System ranks or selects relevant keywords.

Extracted keywords are stored.

Keywords are displayed along with the summary.

4.5.3 Functional Requirements

- FR-52: The system SHALL identify important keywords from lecture content.
- FR-53: The system SHALL identify important concepts relevant to the lecture.
- FR-54: The system SHALL associate extracted keywords with the corresponding lecture.
- FR-55: The system SHALL display extracted keywords with the generated summary.
- FR-56: The system SHOULD avoid unnecessary or frequently occurring generic words.

4.6 Flashcard Generation

4.6.1 Description and Priority

This feature automatically generates question-and-answer flashcards from lecture content or generated summaries.

Priority: High

4.6.2 Stimulus/Response Sequences

User selects a lecture or summary.

User selects Generate Flashcards.

System analyzes the selected content.

Important concepts are identified.

Question-answer pairs are generated.

Generated flashcards are stored.

Flashcards are displayed to the user.

4.6.3 Functional Requirements

- FR-57: The system SHALL generate question-answer flashcards from processed lecture content.
- FR-58: The system SHALL generate flashcards based on important concepts.
- FR-59: Each flashcard SHALL contain a question and corresponding answer.

- FR-60: The system SHALL associate generated flashcards with the corresponding lecture.
- FR-61: The system SHALL store generated flashcards for later retrieval.
- FR-62: The system SHALL display generated flashcards through the user interface.
- FR-63: The system SHALL report an error if flashcard generation fails.

4.7 Flashcard Practice

4.7.1 Description and Priority

This feature allows students to interactively practice the generated flashcards for revision.

Priority: High

4.7.2 Stimulus/Response Sequences

Student opens a flashcard set.

System displays a question.

Student attempts to answer the question.

Student chooses to reveal the answer.

System displays the corresponding answer.

Student proceeds to the next flashcard.

System continues until the flashcard set is completed.

4.7.3 Functional Requirements

- FR-64: The system SHALL allow students to practice generated flashcards.
- FR-65: The system SHALL display one flashcard question at a time.
- FR-66: The system SHALL allow the student to reveal the corresponding answer.
- FR-67: The system SHALL allow navigation between flashcards.
- FR-68: The system SHALL indicate when a flashcard session is completed.
- FR-69: The system SHOULD allow students to restart a flashcard session.

4.8 Lecture and Study Material Retrieval

4.8.1 Description and Priority

This feature allows users to retrieve previously uploaded lectures, transcripts, summaries, keywords, and flashcards.

Priority: Medium

4.8.2 Stimulus/Response Sequences

User opens the lecture history or study-material section.

System retrieves the user's stored content.

System displays available lectures.

User selects a lecture.

System displays the associated transcript, summary, keywords, and flashcards.

4.8.3 Functional Requirements

- FR-70: The system SHALL store processed lecture information for retrieval.
- FR-71: The system SHALL allow authenticated users to view their previously processed lectures.
- FR-72: The system SHALL retrieve the transcript associated with a selected lecture.
- FR-73: The system SHALL retrieve the summary associated with a selected lecture.
- FR-74: The system SHALL retrieve associated keywords and flashcards.
- FR-75: The system SHALL prevent users from accessing another user's private study materials.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- NFR-11: The system SHALL respond to normal user-interface requests within 3 seconds under normal operating conditions.
- NFR-12: The system SHALL provide upload status information for files requiring extended processing.
- NFR-13: The system SHALL support multiple users simultaneously without loss or corruption of stored data.
- NFR-14: The system SHOULD process uploaded content asynchronously when processing requires significant computational time.
- NFR-15: Database queries for previously stored summaries and flashcards SHOULD normally complete within 3 seconds.
- NFR-16: The system SHALL provide appropriate feedback when transcription, summarization, or flashcard generation is still in progress.

5.2 Safety Requirements

- NFR-17: The system SHALL prevent malformed or unsupported files from being processed.
- NFR-18: The system SHALL prevent system failure caused by excessively large or invalid user inputs.
- NFR-19: The system SHALL handle processing failures without corrupting existing user data.
- NFR-20: The system SHALL provide appropriate error messages when processing cannot be completed.
- NFR-21: Temporary files created during processing SHOULD be removed after they are no longer required.

5.3 Security Requirements

- NFR-22: The system SHALL authenticate users before allowing access to protected resources.
- NFR-23: The system SHALL enforce role-based authorization for students, educators, and administrators.
- NFR-24: All client-server communication SHALL use HTTPS with TLS 1.2 or higher.
- NFR-25: User passwords SHALL be stored using secure one-way password hashing.

- NFR-26: The system SHALL prevent unauthorized access to user lecture files, transcripts, summaries, and flashcards.
- NFR-27: The system SHALL validate uploaded files before storing or processing them.
- NFR-28: The system SHALL validate and sanitize user-provided input to reduce security vulnerabilities.
- NFR-29: Administrative functions SHALL be accessible only to authenticated administrators.
- NFR-30: The system SHALL comply with applicable data privacy requirements.
- NFR-31: Sensitive information SHALL not be unnecessarily exposed in API responses or error messages.

5.4 Software Quality Attributes

- NFR-32: The system SHALL provide a simple and intuitive interface suitable for students and educators.
- NFR-33: Common operations such as uploading lectures, viewing summaries, and practicing flashcards SHOULD require minimal navigation.
- NFR-34: The system SHALL handle processing errors gracefully without terminating the entire application.
- NFR-35: Stored lecture data SHALL remain consistent after successful processing.
- NFR-36: The backend SHALL use a modular architecture separating API, processing, authentication, and database components.
- NFR-37: The frontend SHALL use reusable React components where appropriate.
- NFR-38: Source code SHALL follow consistent naming and documentation practices.
- NFR-39: The system SHALL be designed to support an increasing number of users and lecture-processing requests.
- NFR-40: The processing architecture SHOULD allow computationally intensive tasks to be handled independently from normal API requests.
- NFR-41: The client application SHALL operate on supported modern browsers.
- NFR-42: The server application SHALL be deployable on a Linux-based environment.
- NFR-43: Major system components SHALL be independently testable.
- NFR-44: API endpoints SHALL support automated functional testing.

5.5 Business Rules

- BR-01: Only authenticated users SHALL be permitted to upload and access lecture materials.
- BR-02: Students SHALL have access to their own lectures, summaries, and flashcards.
- BR-03: Educators SHALL be permitted to upload lecture content and generate or share flashcards.
- BR-04: Administrators SHALL have privileges to manage user accounts.
- BR-05: Users SHALL only be allowed to perform operations permitted by their assigned role.
- BR-06: A lecture SHALL be processed only after successful file validation.
- BR-07: Flashcards SHALL be generated only when valid lecture content or a summary is available.
- BR-08: A generated transcript SHALL be associated with the lecture from which it was produced.
- BR-09: A generated summary SHALL be associated with its corresponding transcript or lecture.
- BR-10: Generated flashcards SHALL be associated with the corresponding lecture or summary.

6. Other Requirements

6.1 Database Requirements

- DR-01: The system SHALL use MongoDB Atlas for persistent data storage.
- DR-02: The database SHALL store user account information required for authentication and authorization.
- DR-03: The database SHALL store lecture metadata.
- DR-04: The database SHALL store generated transcripts.
- DR-05: The database SHALL store generated summaries.
- DR-06: The database SHALL store extracted keywords and concepts.
- DR-07: The database SHALL store generated flashcards.
- DR-08: Relationships between users, lectures, transcripts, summaries, and flashcards SHALL be maintained using appropriate MongoDB document structures and references.
- DR-09: Database access SHALL be restricted to authorized backend components.
- DR-10: Regular database backup SHOULD be implemented to reduce the risk of permanent data loss.

6.2 Data Retention Requirements

- DR-11: The system SHALL retain successfully processed lecture metadata and generated study materials according to the application's data-retention policy.
- DR-12: Temporary processing files SHOULD be deleted when they are no longer required.
- DR-13: Users SHOULD be provided with an option to remove their stored lecture materials where applicable.

6.3 Legal and Privacy Requirements

- LR-01: The system SHALL handle user data in accordance with applicable data privacy regulations.
- LR-02: The system SHALL inform users about the collection and processing of their uploaded lecture content.
- LR-03: User data SHALL not be disclosed to unauthorized parties.
- LR-04: Third-party transcription or cloud services SHALL be used in accordance with their applicable terms and privacy policies.

6.4 Internationalization

- IR-01: The initial version of the system SHALL primarily support English-language lecture content.
- IR-02: The system SHOULD be designed so that additional languages can be supported in future versions.
- IR-03: User-interface text SHOULD be structured so that localization can be added without major architectural changes.

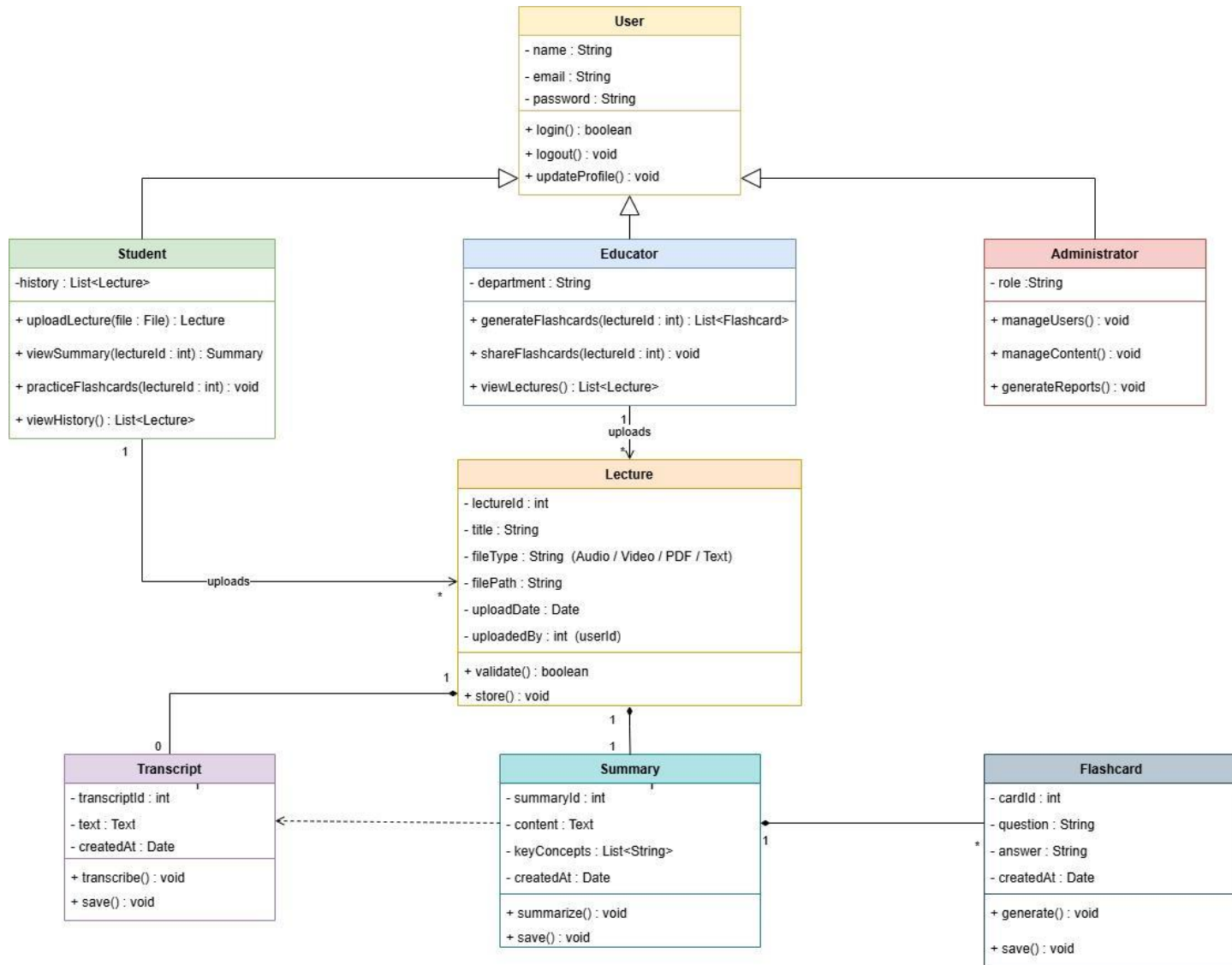
Appendix A: Glossary

Term	Definition
API	Application Programming Interface used for communication between software components.
AI	Artificial Intelligence, techniques used to perform tasks that normally require human intelligence.
Backend	Server-side part of the application responsible for processing requests and data.
Database	Organized collection of data stored and managed electronically.
FastAPI	Python web framework used to develop the backend APIs.
Flashcard	A question-and-answer based study material used for revision.
Frontend	User-facing part of the application with which users interact.
HTTPS	Secure version of HTTP used for encrypted communication.
Keyword	An important word or term representing a major topic in lecture content.
NLP	Natural Language Processing, techniques for processing and understanding human language.
PDF	Portable Document Format used for digital documents.
MongoDB Atlas	A cloud-based database service that provides managed MongoDB databases for storing and retrieving application data.
REST API	Web API architecture that uses HTTP methods to communicate between applications.
React.js	JavaScript library used for developing user interfaces.

Summary	A concise representation of the important information in lecture content.
Transcript	Text representation of spoken lecture content.
TLS	Transport Layer Security protocol used to secure network communication.
UML	Unified Modeling Language used for visualizing software systems.
Whisper	Speech-to-text model used to convert spoken audio into text.

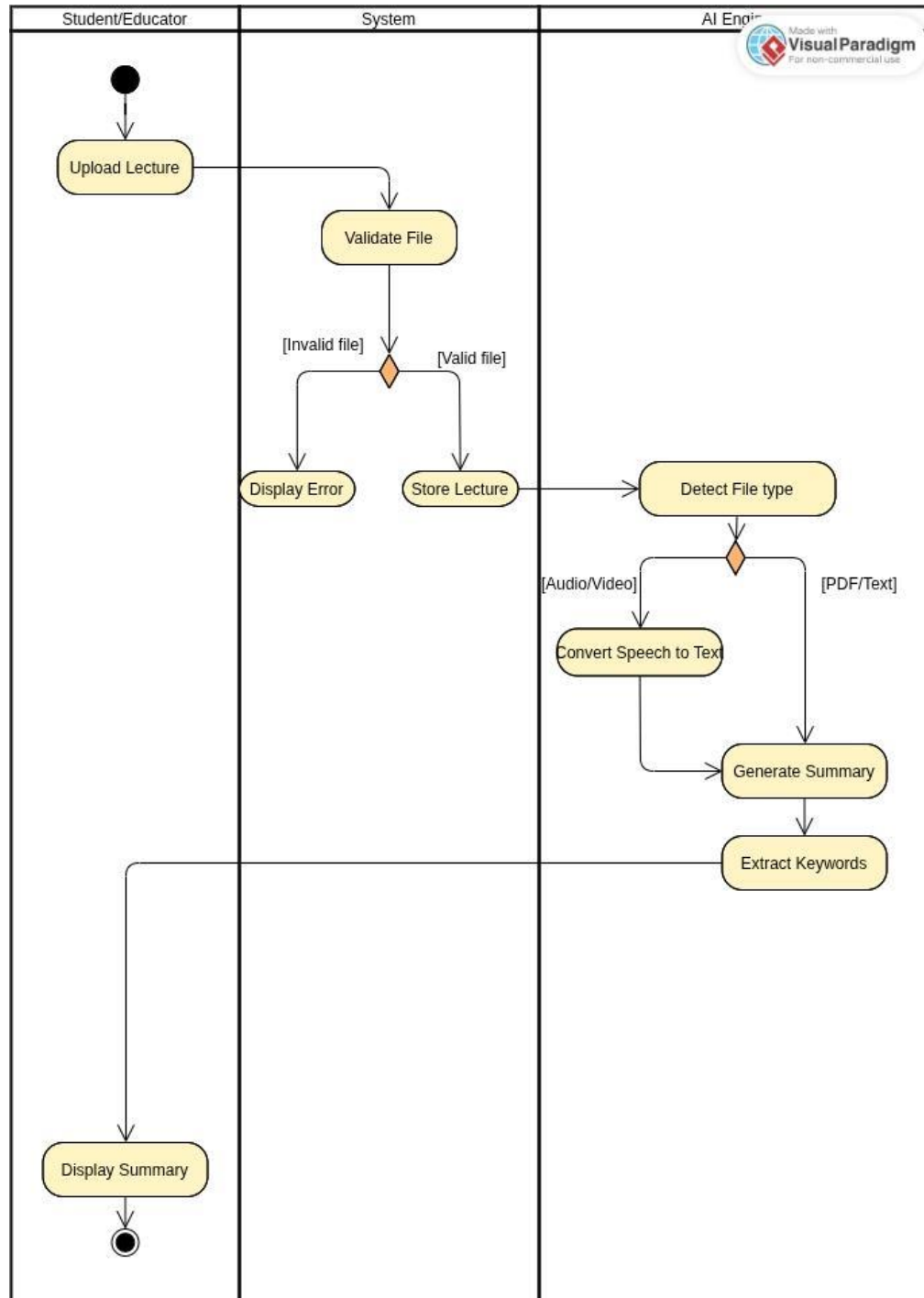
Appendix B: Analysis Models

B.1 UML Class Diagram

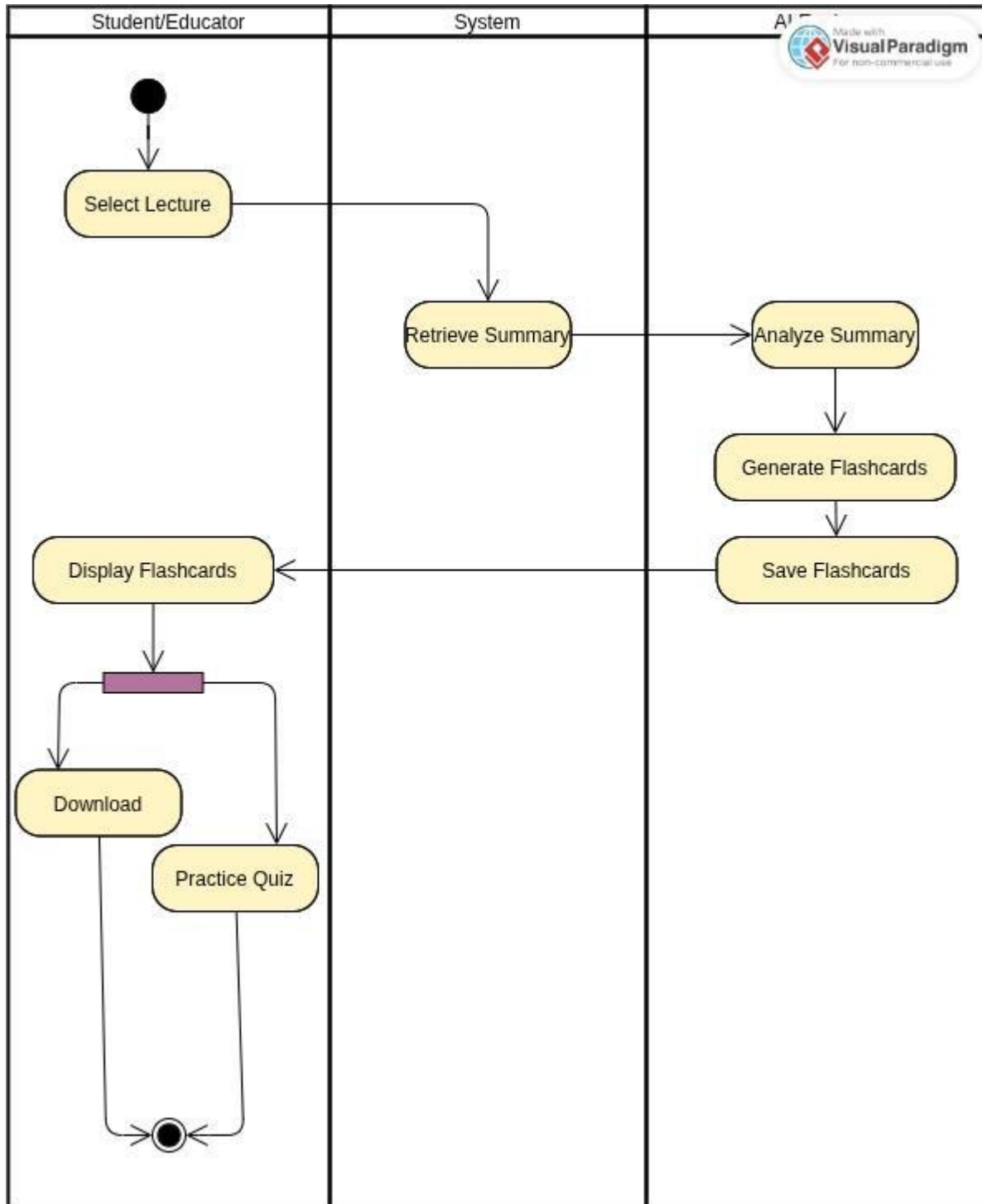


B.2 UML Activity Diagram

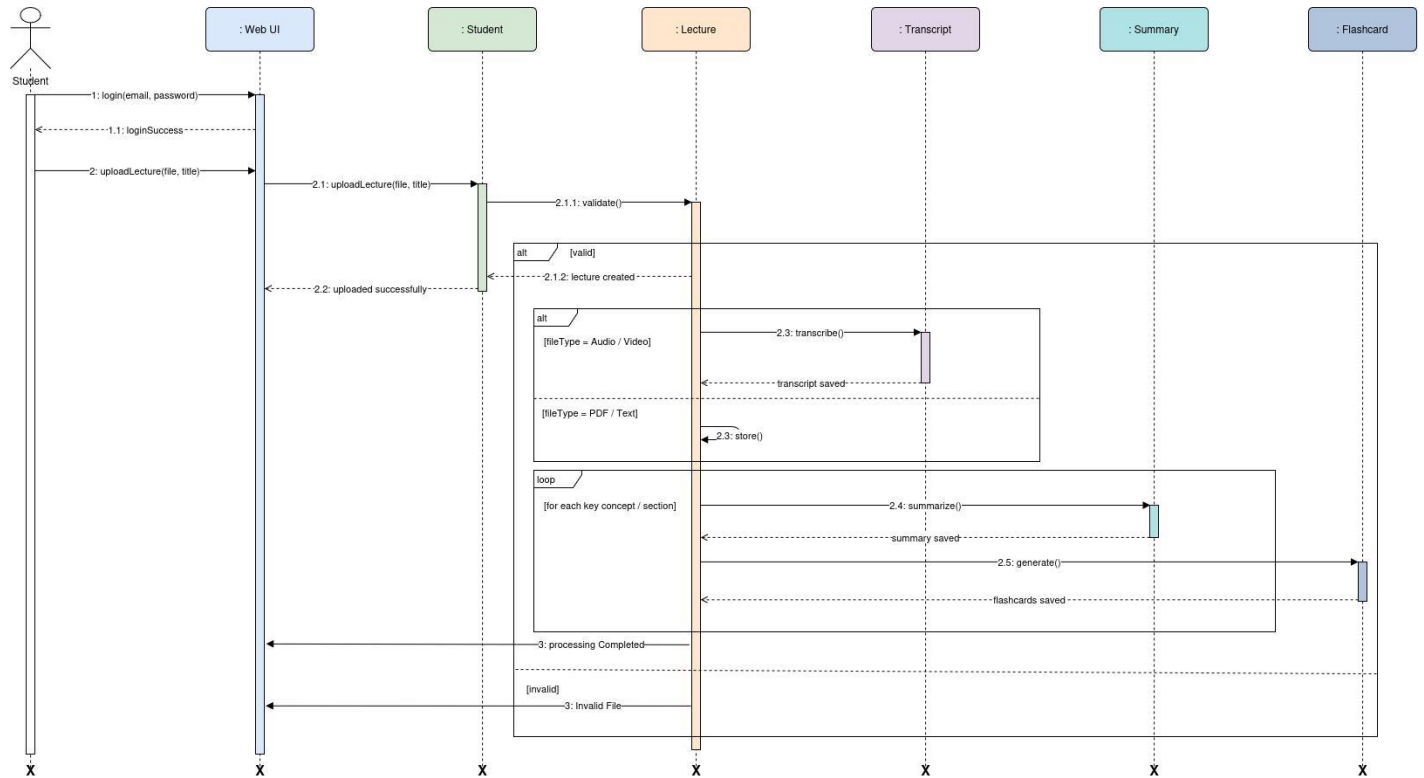
Activity Diagram 1: Upload Lecture and Generate Summary



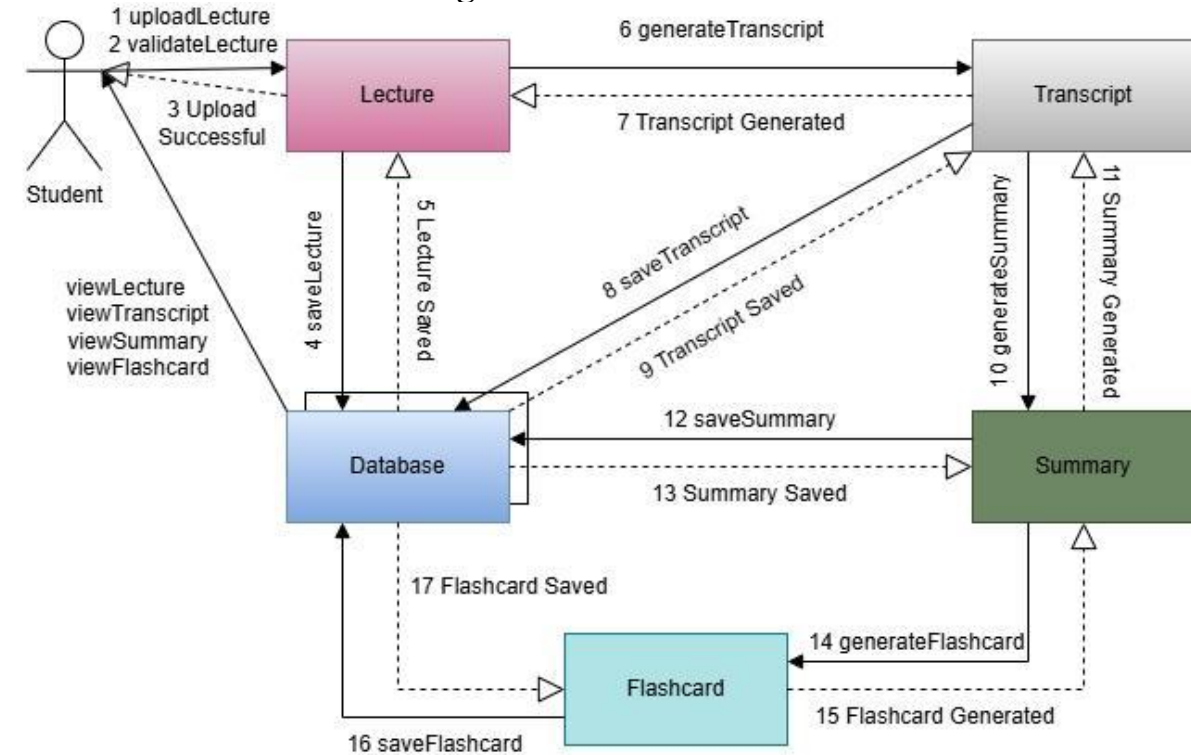
Activity Diagram 2: Generate and View Flashcards



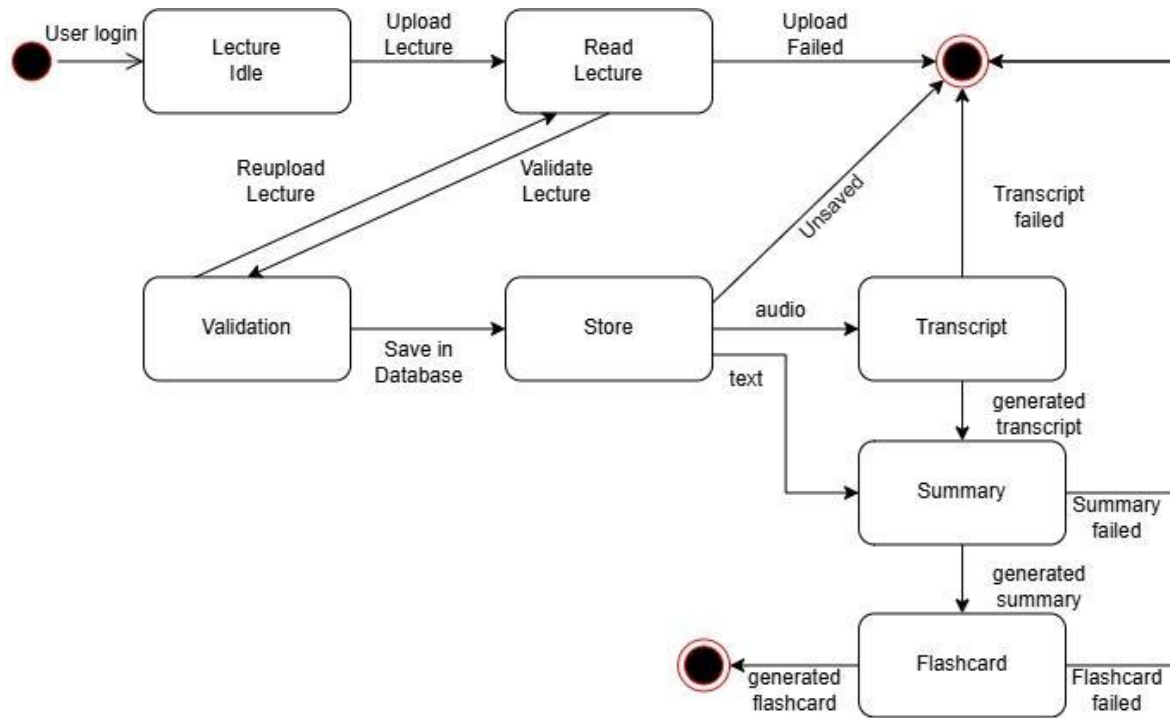
B.3 UML Sequence Diagram



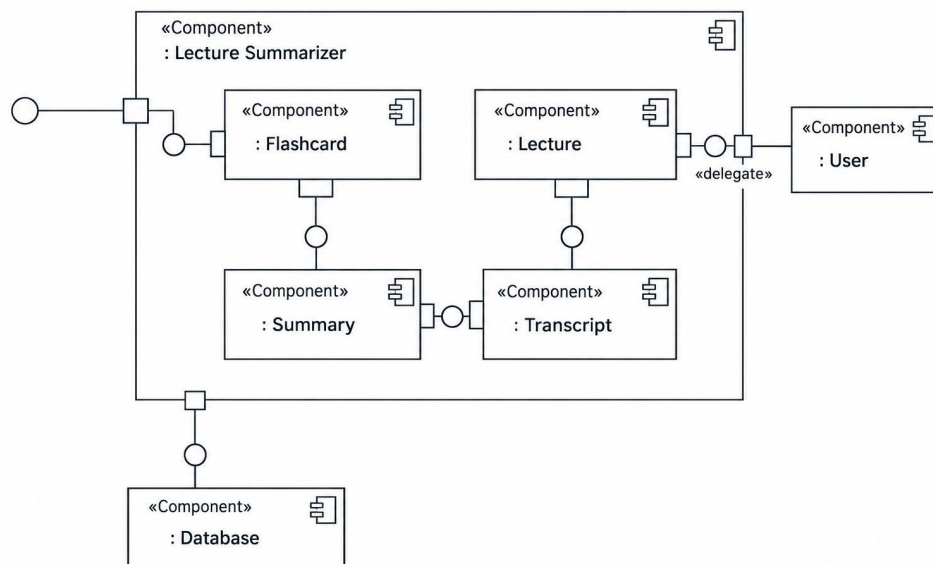
B.4 UML Collaboration Diagram



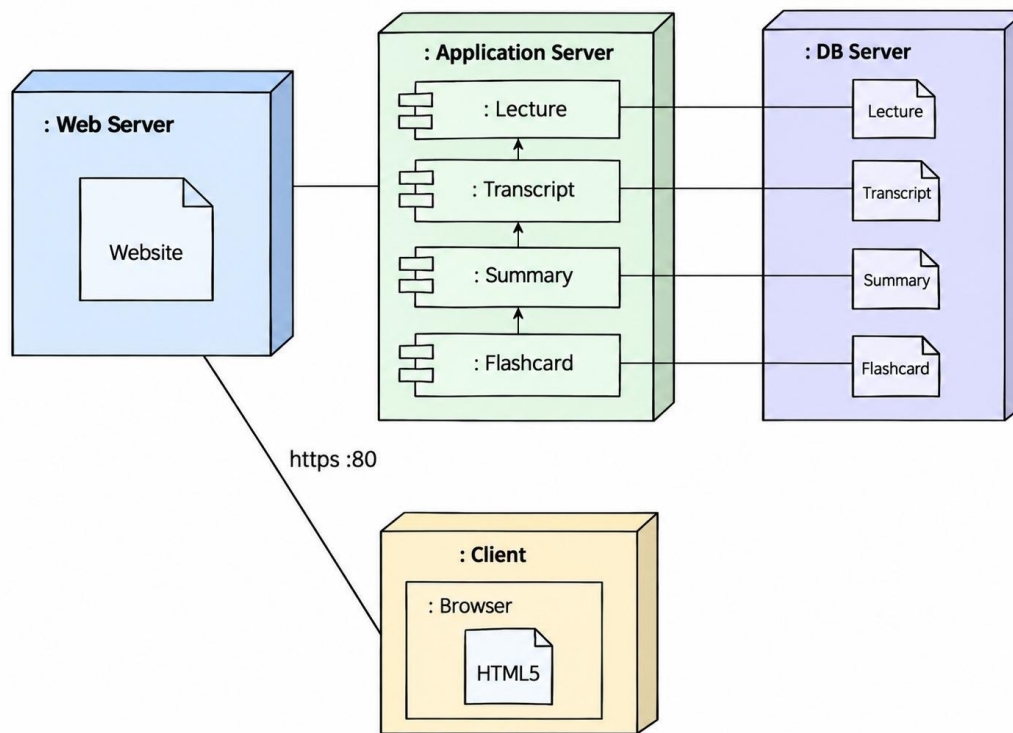
B.5 UML State Chart Diagram



B.6 UML Component Diagram



B.7 UML Deployment Diagram

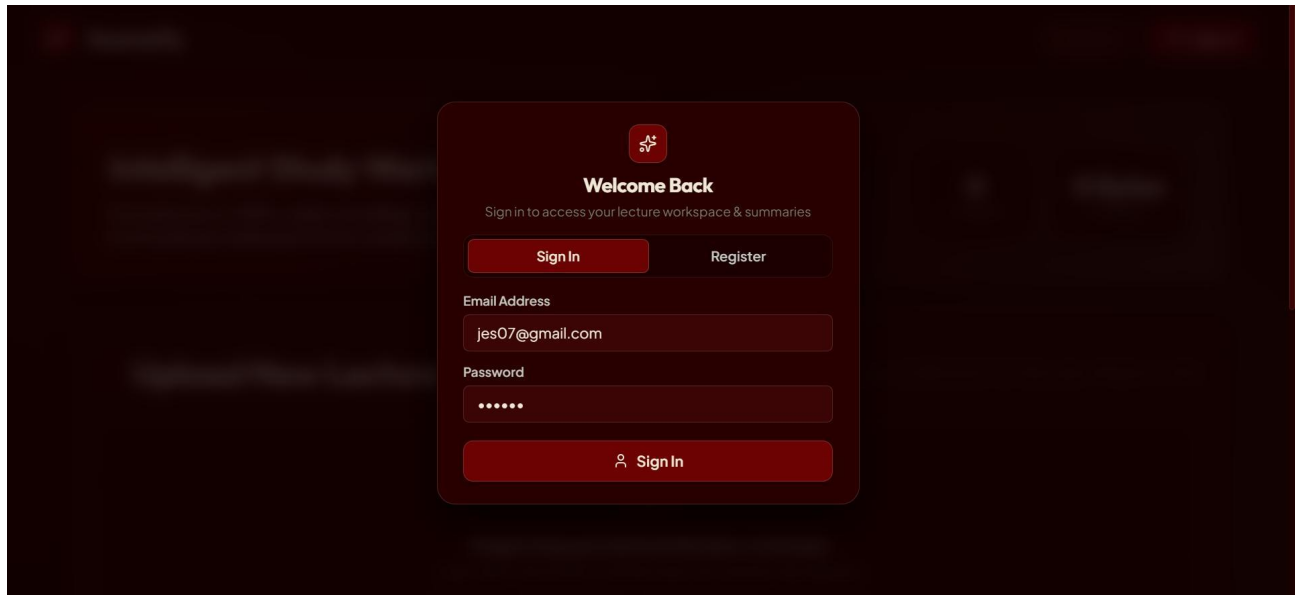


Appendix C: To Be Determined List

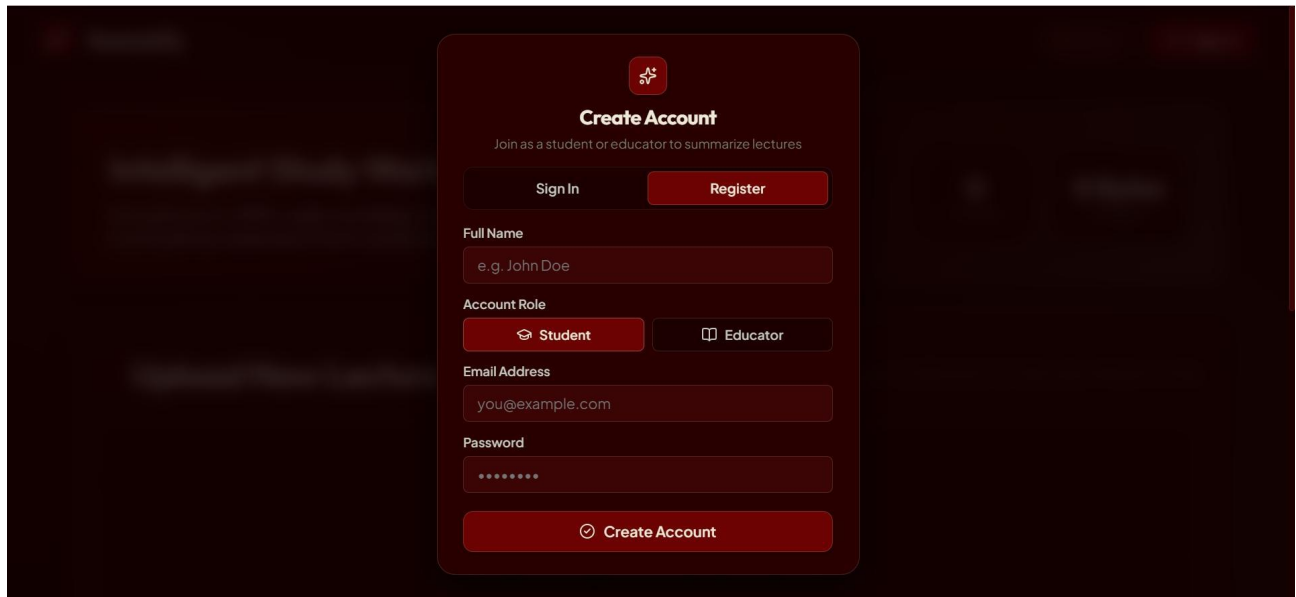
- TBD-01: Maximum permitted lecture file size.
- TBD-02: Exact list of supported audio and video formats.
- TBD-03: Exact maximum duration of an uploaded lecture.
- TBD-04: Final NLP model to be used for summarization.
- TBD-05: Final flashcard-generation model or algorithm.
- TBD-06: Exact number of flashcards generated per lecture.
- TBD-07: Final cloud-storage provider, if required.
- TBD-08: Final deployment infrastructure and server specifications.
- TBD-09: Final authentication mechanism and session-management strategy.
- TBD-10: Final data-retention period for uploaded lecture files.
- TBD-11: Final supported languages for lecture processing.
- TBD-12: Exact performance benchmarks for transcription and summarization based on lecture duration and server configuration.

APPLICATION SCREENSHOTS

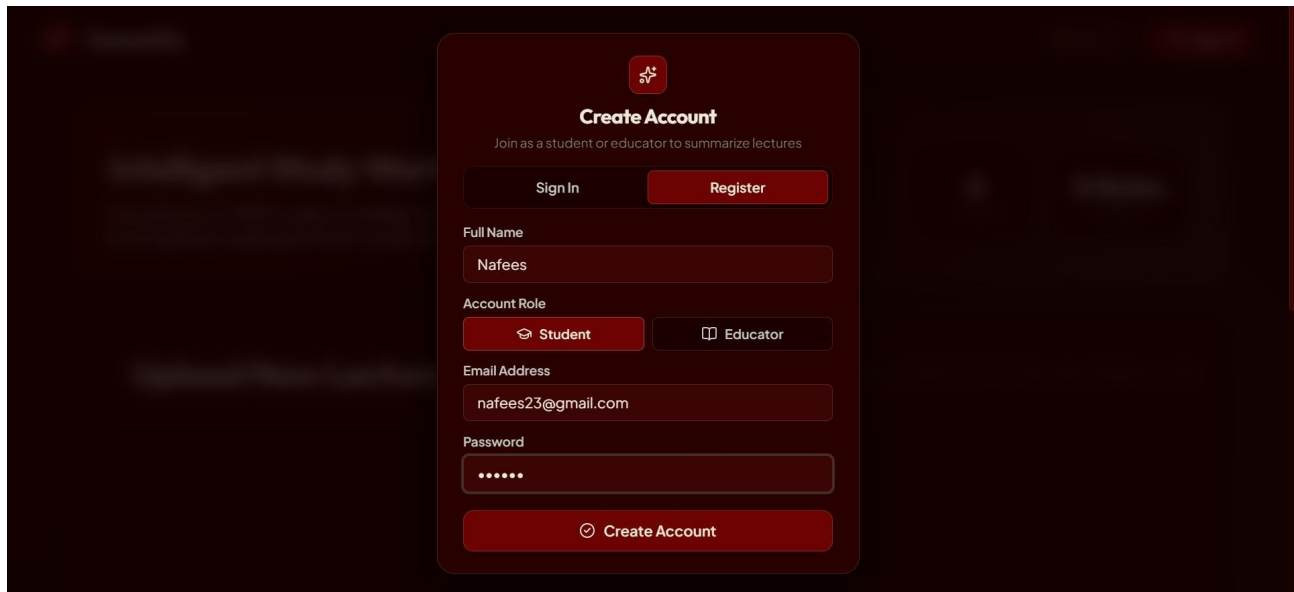
Sign in page



Register page



Student user



The screenshot shows a 'Create Account' form for a student user. The form is titled 'Create Account' with a subtitle 'Join as a student or educator to summarize lectures'. It features a 'Sign In' button and a 'Register' button. The form fields are: Full Name (Nafees), Account Role (Student, Educator), Email Address (nafees23@gmail.com), and Password (*****). A 'Create Account' button is at the bottom.

Create Account
Join as a student or educator to summarize lectures

[Sign In](#) [Register](#)

Full Name
Nafees

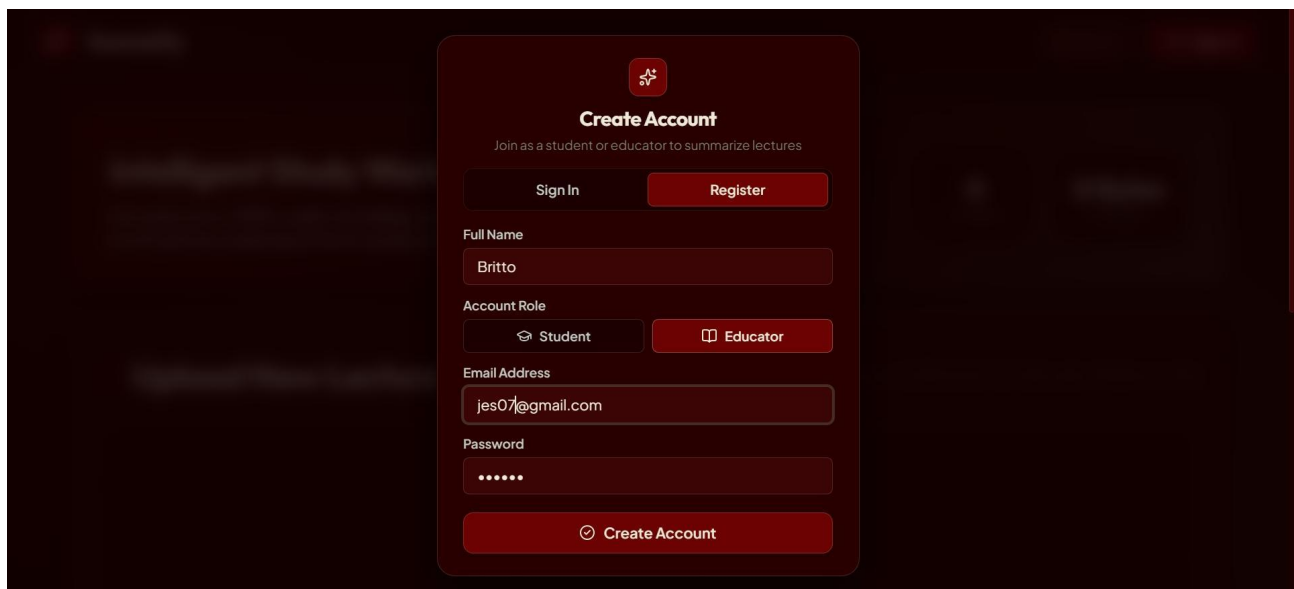
Account Role
☒ Student ☐ Educator

Email Address
nafees23@gmail.com

Password

[Create Account](#)

Educator user



The screenshot shows a 'Create Account' form for an educator user. The form is titled 'Create Account' with a subtitle 'Join as a student or educator to summarize lectures'. It features a 'Sign In' button and a 'Register' button. The form fields are: Full Name (Britto), Account Role (Student, Educator), Email Address (jes07@gmail.com), and Password (*****). A 'Create Account' button is at the bottom.

Create Account
Join as a student or educator to summarize lectures

[Sign In](#) [Register](#)

Full Name
Britto

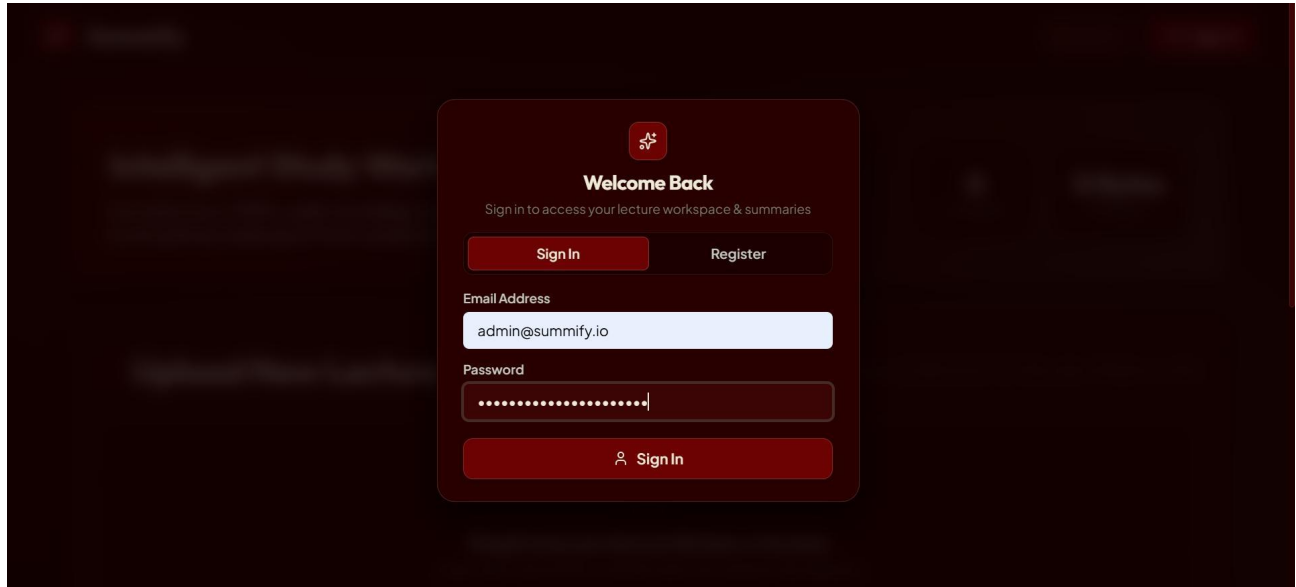
Account Role
☐ Student ☒ Educator

Email Address
jes07@gmail.com

Password

[Create Account](#)

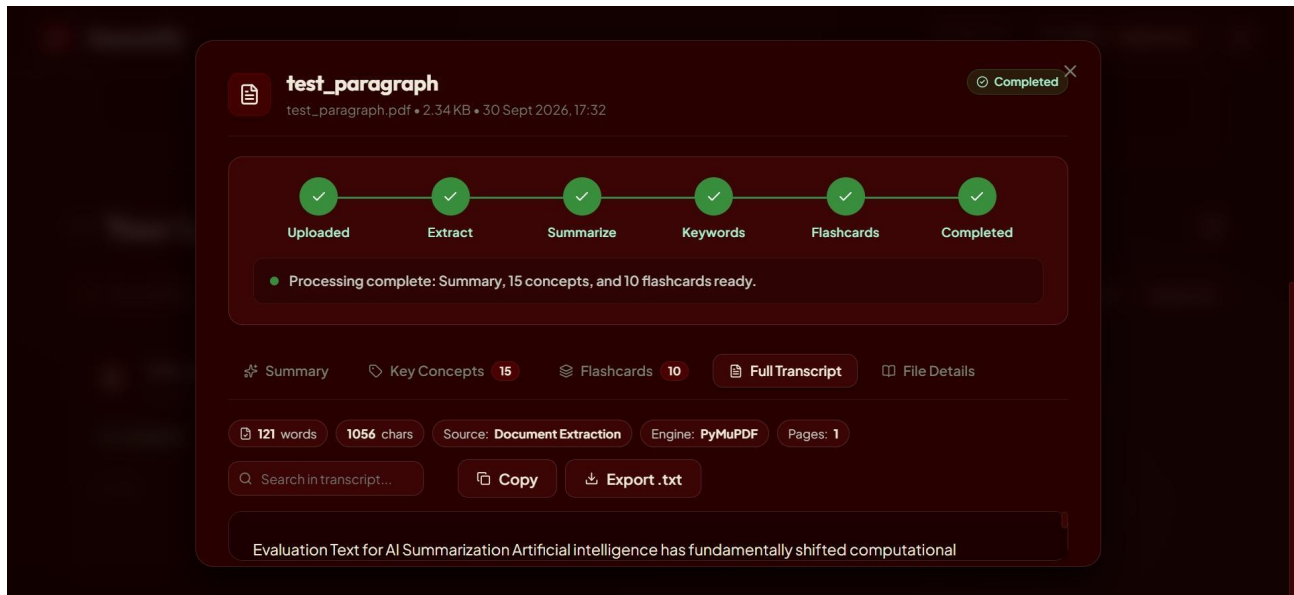
Administrator user



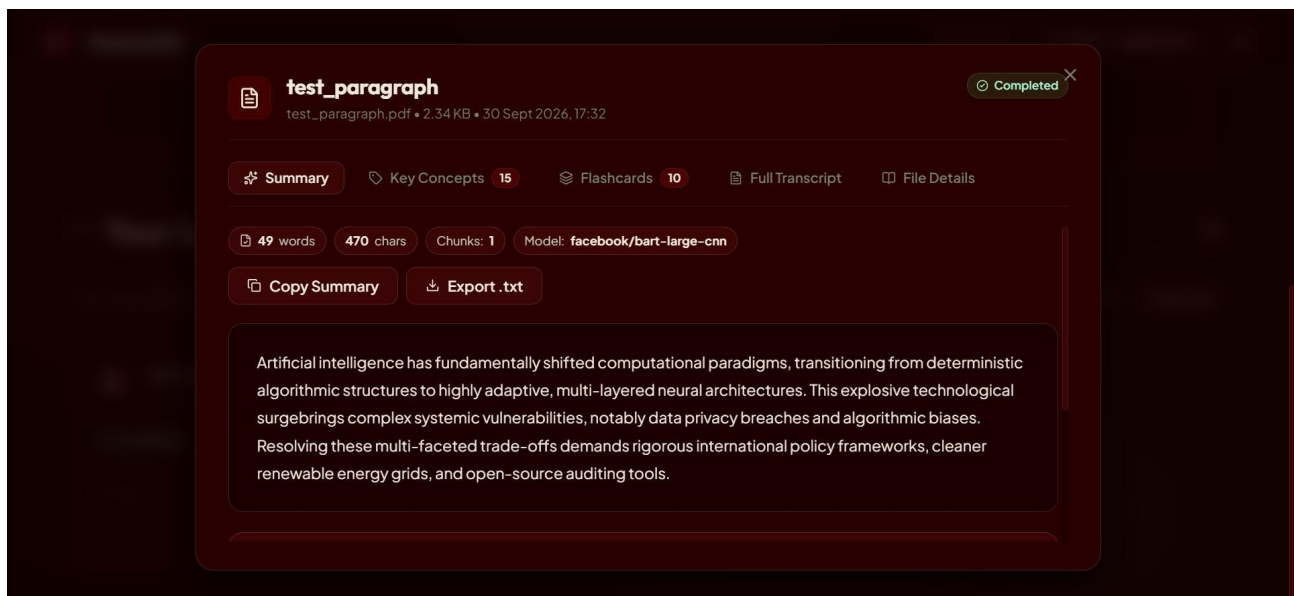
Upload Lecture



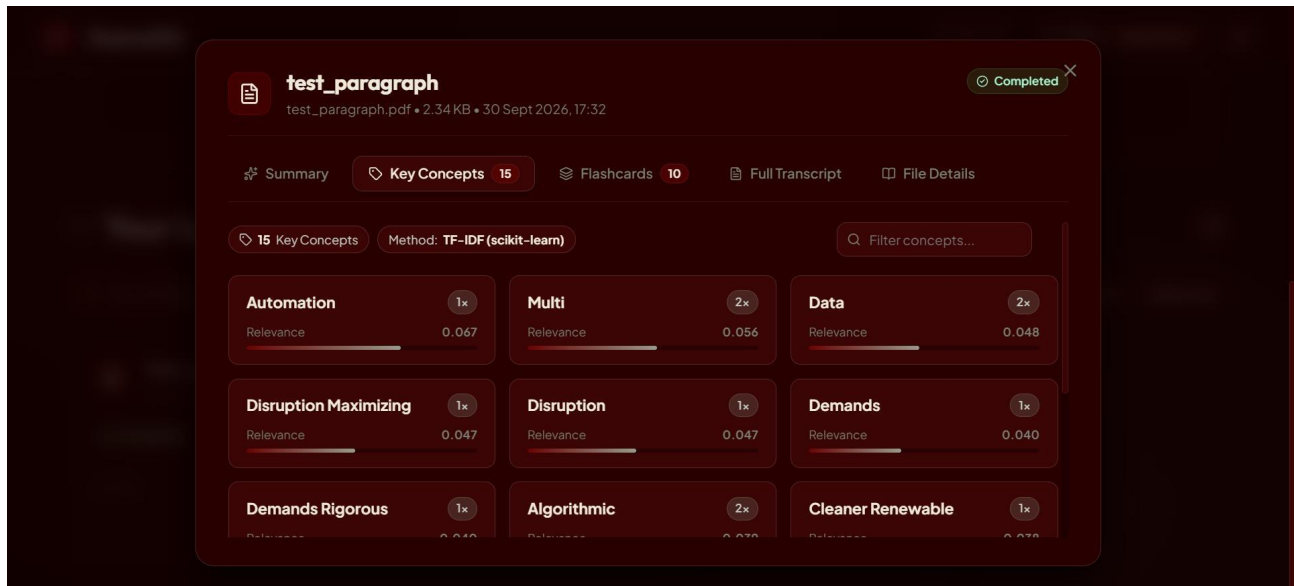
Processing Pipeline



Summary

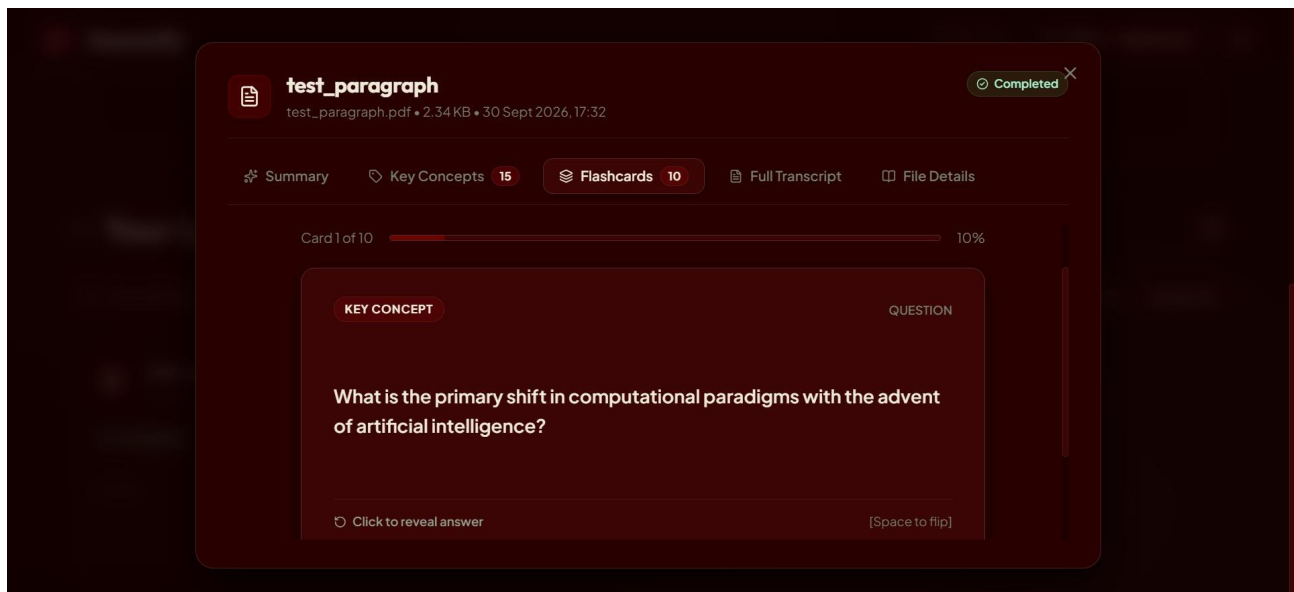


Key Concepts

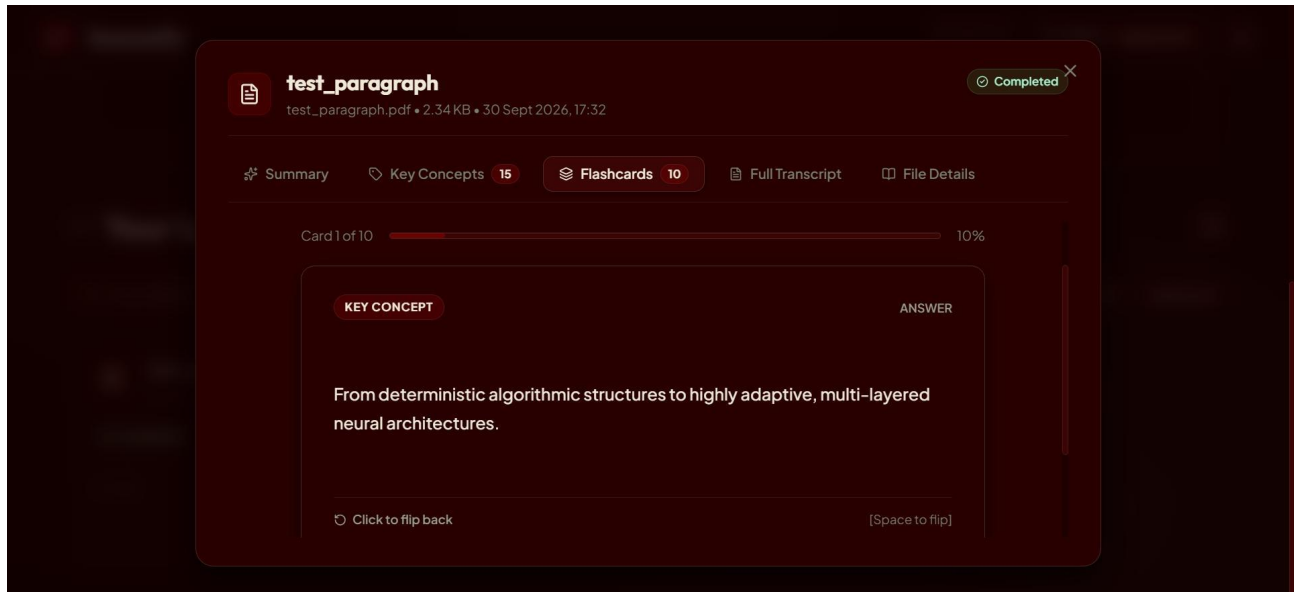


Flashcard (Study mode)

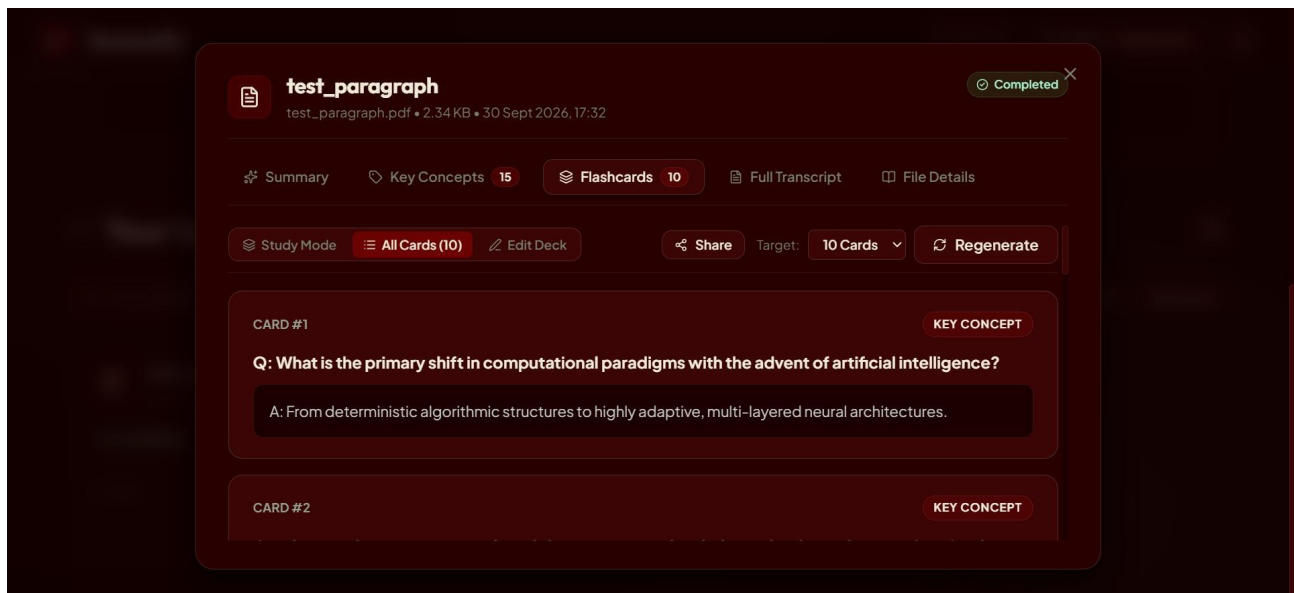
Question:



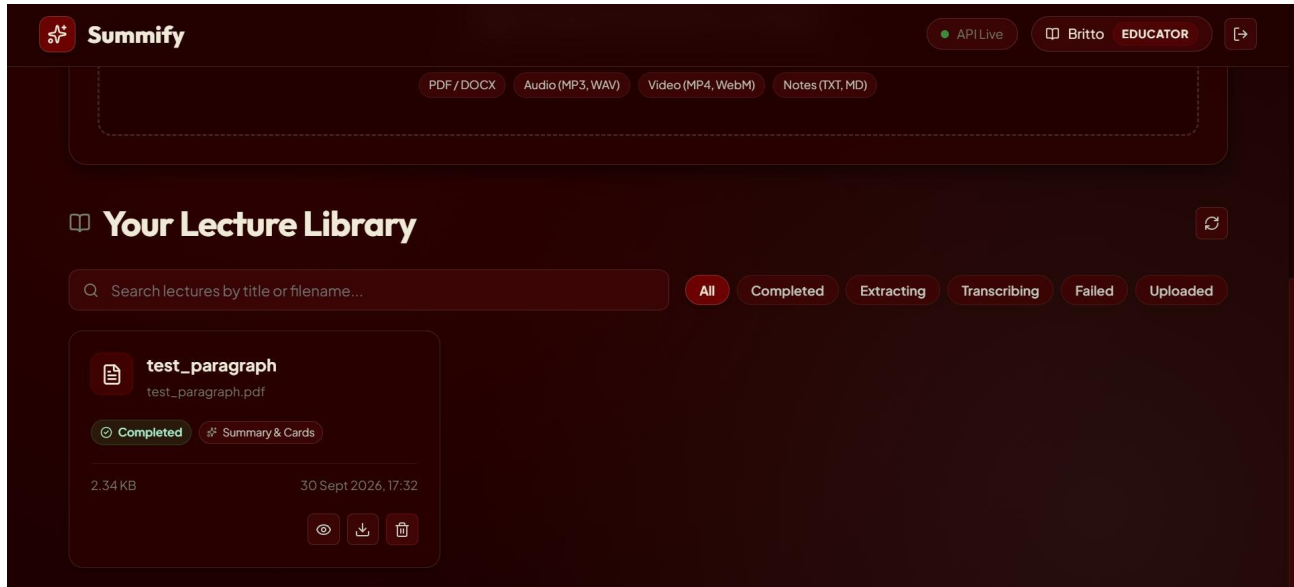
Answer:



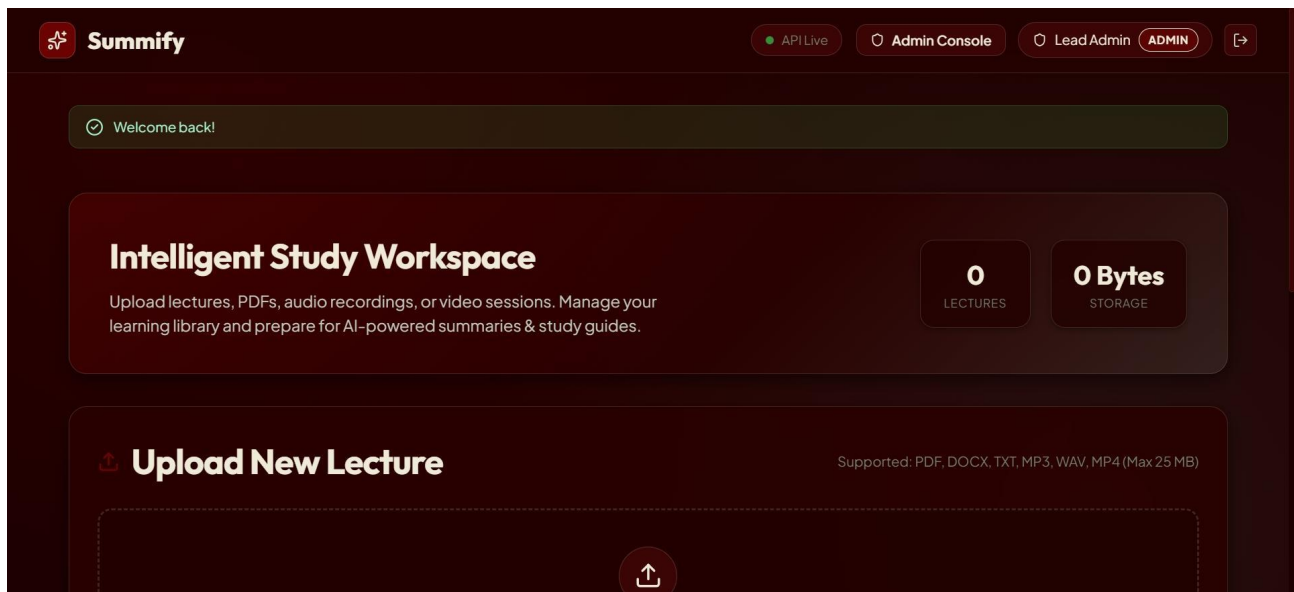
Flashcards (All cards)



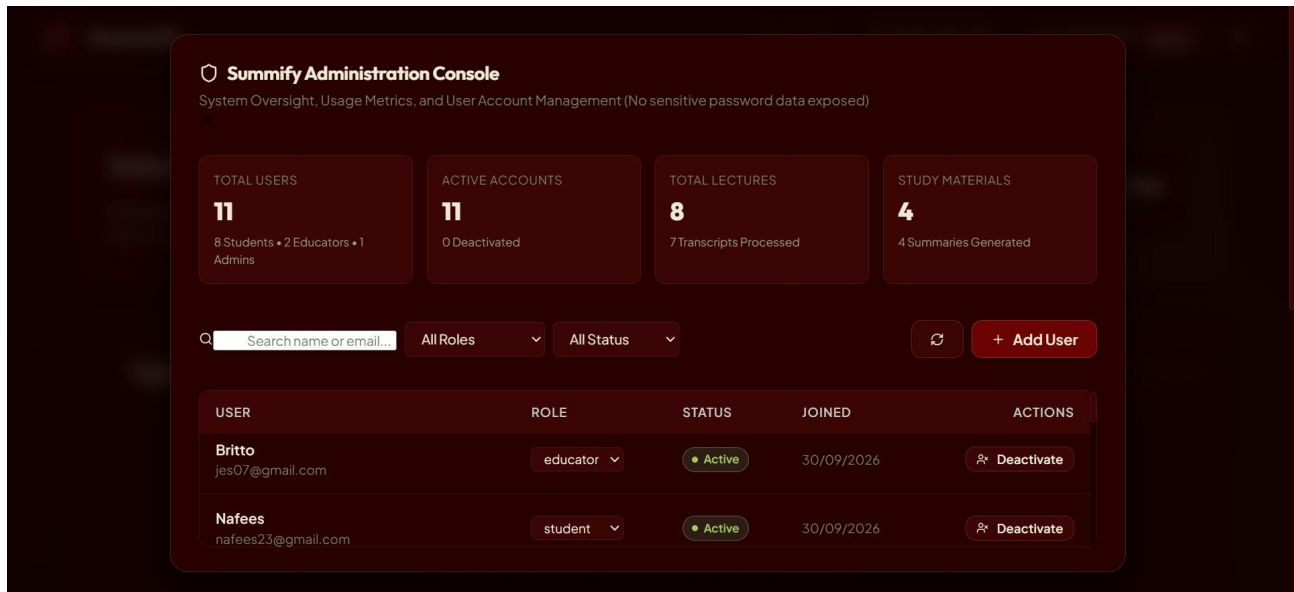
Lecture History



Admin page



Administrator Console



E-RESOURCE

Github link: <https://github.com/Shreehari-V/Summify>